

GENG5505 — Question Bank (Corrected against Lecture Slides)

本文档基于原 5505.docx 题库，结合 Week 1–Week 12（无 Week 8）共 20 个 lecture PDF 中的内容进行校对修正。

凡是与 lecture 内容不一致、措辞不准确、术语错误或回答有遗漏的地方，均在原答案的基础上修订并补充。

每个被修正的题目下方均附 △ **修正说明 (Correction)**，标明依据的具体 lecture。

原始 5505.docx、Week 文件夹及任何其他文件均未改动。

Section A

Question 1 — Explain the 4 stages of the project life cycle and why each is important

The 4 stages (per Hartley 2018, lectured in WK1b):

Concept → Planning → Execution (monitoring & controlling) → Finalisation (and evaluation).

Stage 1 — Concept Stage (*also called initiation, feasibility, or Stage 1*)

The **idea stage** in which the project is conceived. The team discusses preliminary goals, deliverables and strategic vision alignment; raises problems; conducts impact assessment in terms of **TBL and life cycle**; identifies potential benefits; researches alternative approaches; and determines provisional costing.

Why it is important. One of the most common causes of project failure is miscommunication, misunderstanding, and lack of clarity between stakeholder expectations and the project's final output. Stakeholders often ignore the basic principle of project management — *identify the problem before you try to solve it*. The concept stage enforces this discipline because it requires the project to be correctly **justified, investigated, assessed and approved**. It captures, documents and clarifies relevant information on the **four input constraints** found in every project: **Time, Cost, Resources and Specification** (Hartley 2018, WK1b).

Example (PCH). Initial purpose: replace the old rundown children's hospital with a new tertiary one; preliminary economic, social and environmental issues raised; impact assessment performed; budget approximated at \$1.2 billion; site set in Nedlands; ~290 beds; bright child-friendly design.

Stage 2 — Planning Stage (*also called schedule, preparation, or Stage 2*)

Following the decision to proceed in Stage 1, all work required for successful completion is planned and scheduled. **Objectives are finalised, resources are assigned, quality is signed off (including TBL and life-cycle guidelines & standards), final costs are approved, timing is agreed, and all other administrative matters are determined** (WK1b).

Why it is important. Without a detailed plan, execution has no direction, resource alignment, or budget control. The planning stage prevents costly surprises and is where stakeholder alignment is consolidated — misalignment caught here is far cheaper to fix than misalignment caught during construction.

Example (PCH). Design detailed further; budget \$1.2B funded by WA government; resources allocated; timeline outlining each construction phase, projected opening June 2015.

Stage 3 — Execution Stage (*also called implementation, control, or Stage 3*)

The project has commenced. Emphasis shifts to **tracking actual progress against the schedules developed in Stage 2 as the comparison point of reference**. All work (including TBL & life-cycle thinking) is **monitored, controlled and corrected** where necessary; schedules are reviewed, revised and updated as required (WK1b).

Why it is important. This is where the majority of project expenditure occurs; poor execution directly produces cost over-runs, delays and quality failures. Monitoring & control mechanisms allow early detection and correction of deviations before they compound.

Example (PCH). Physical construction commenced; delays occurred due to asbestos and water contamination; timelines repeatedly reviewed, revised and updated.

Stage 4 — Finalisation (and Evaluation) Stage (*also called completion, termination, hand-over, or Stage 4*)

The project has been completed and the deliverable handed over to the client. **Resources are disposed of or reassigned, the project is evaluated, reports are written and presented, and the administrative arm of the project is closed** (WK1b).

Why it is important. Project success cannot be measured solely against technical scope; the finalisation stage formally evaluates whether the project achieved its strategic purpose and stakeholder value. Lessons learned are documented so knowledge is retained for future projects. Formal hand-over establishes clear accountability between contractor and client.

Example (PCH). Formal acceptance & hand-over of the new PCH from John Holland to the WA Government; numerous government and consulting evaluation reports published.

⚠ **Correction (vs. original docx):** Original answer was largely correct but used the more colloquial *"is to handover to the client. Lesson learnt"*. The lecture (WK1b, "Project Management Lifecycle" slide) uses the precise wording **"Finalisation (and evaluation) stage — also called completion, termination, hand-over, or Stage 4"**, with deliverables being **handed over, resources disposed of or reassigned, the project evaluated, reports written and presented, and the administration arm of the project closed**. Added the four input constraints (Time, Cost, Resources, Specification) which the original answer omitted.

Question 2 — Explain "sustainable approach", how it is consistent with TBL, and how applying TBL to every stage improves the project

A "sustainable approach", as defined in lecture (WK1a, Robbins et al. 2009)

"A company's ability to achieve its business goals and increase **long-term shareholder value** by integrating economic, environmental and social opportunities into its business strategies — the **Triple Bottom Line (TBL)** approach."

And, more fundamentally:

"Sustainability — **meeting the needs of people today without compromising the ability of future generations to meet their own needs.**"

The key insight from the slides: the long-term financial interests of a company are **not** mutually exclusive with acting fairly in the interests of stakeholders.

Indicators of sustainability — consistent with TBL (Labuschagne et al. 2005, WK1a)

Dimension	Indicator examples	Guiding question
Economic / Prosperity	Financial health, economic performance, ROI, ROA, risk	Will the project generate prosperity & enhance the affected economies?
Environmental / Planet	Mineral & energy resources, air, water & land resources, emissions, waste, hazardous materials	Will the project cause long-term damage to the environment?
Social / People	Internal HR, external population, community, stakeholder participation	Will the project be implemented in a socially responsible manner & benefit the affected communities fairly & equitably?

TBL applied across the 4 stages

Concept — Embed sustainability at the very point the idea is justified, so that the project is approved on TBL-aligned grounds (economic catalyst + social benefit + environmental constraints identified). PCH example: localising procurement to John Holland; tailoring spaces for children (social); \$1.2B economic injection (economic); water-recycling and LED strategies (environmental).

Planning — TBL is built into risk identification, scope, contracts, quality standards and approval gates. Design choices in PCH such as the double-skinned façade (long-run energy saving) and bright play areas (social) were locked in as planning decisions.

Execution — TBL becomes an active management tool: studies confirm no significant environmental impact; sustainable choices conserve labour, natural resources and energy; the resulting IP and reputation produce long-term returns for stakeholders.

Finalisation — Project success cannot be measured by scope delivery alone; further analysis using TBL is required. **Quality checklists** are useful tools to ensure TBL is considered during quality analysis. A company giving regular attention to sustainability earns a strong positive public image (e.g. PCH's community engagement and outreach to vulnerable populations / low-SES communities).

Benefits of life cycle thinking (WK1b)

- Savings in labour, resources & energy
- Positive image → competitive advantage
- Intellectual property & long-term returns
- Alignment with TBL

⚠ **Correction:** Original answer mixed up the *measurement question* for each TBL dimension (e.g. "will the project generate prosperity") — these are quoted directly from Keeble et al. 2003 in WK4b. Also clarified the formal definitions of "sustainable approach" and "sustainability" as Robbins et al. 2009 (WK1a). The original phrasing "*with specialist identified the possible risks*" in the Planning section was kept as a PCH example but I have made the underlying TBL principle explicit.

Section B

B1 — Differentiate the 3 numerical project-selection methods; which is the most exhaustive?

The lecture (WK2a, "Project selection: Numeric models") lists three numeric methods:

1. **Payback Period** — the time required to recover the initial investment.
Formula: $\text{Payback} = \text{Cost of project} \div \text{Annual cash revenue}$.
Simple, but ignores cash flow after the payback point and the time value of money.
2. **Return on Investment (ROI)** — the overall profit on an investment expressed as a percentage of total investment.
Formula: $\text{ROI} = (\text{Net profit} \div \text{Total investment}) \times 100$.
Example from the slide: net profit 40,000 over 6 years on an investment of 100,000 $\rightarrow$ ROI = 40%.
Considers the entire cash-flow period, **but does not account for the time value of money**.
3. **Net Present Value (NPV) — the most exhaustive** — the projected profitability of an investment based on **future (anticipated) cash-flows discounted (from year 2) at a stated interest rate**.
 $\text{NPV} = \sum (\text{Cash flow} \times \text{Discount factor})$, where $\text{Discount factor} = 1 / (1 + i)^n$.
Most exhaustive because it accounts for both the entire cash-flow period **and** the time value of money.

⚠ **Correction:** Original answer's ROI formula and Payback formula were correct but missing context. Added the key distinction that **ROI ignores time value of money** but covers the whole cash-flow period — which is why NPV is the most exhaustive. NPV formula clarified per WK2a slide.

B2 — Stages of team development (Tuckman, lectured in WK5a)

Team development is **not a linear process** — teams can regress to earlier stages. The five stages are:

1. **Forming** — Everyone comes together for the first time; introductions, attempts to "break the ice"; tentative relationships; little trust; unknown goals; ambiguity in goal and role; hidden agendas; "cards held close to the chest." (*Group example: first class meeting.*)
2. **Storming** — Inevitable and natural conflict appears. Caused by personality clashes, team-role disputes, leadership issues, or any other aspect of the team's processes. (*Meetings 2–3, no clear direction on project progress.*)

3. **Norming** — The "unwritten" rules, codes of conduct and acceptable behaviours (norms) are developed, shared and agreed. Once established, all members know what to expect from each other. (*Regular meetings with updates.*)
4. **Performing** — Team members are ready to perform their assigned work in a united, cohesive and productive manner. (*Group actively contributing to the report.*)
5. **Adjourning** — The project has finished and the team members disband. (*Project submitted, final presentation completed.*)

⚠ **Correction:** Original answer spelled stage 4 as "**Preforming**" — should be "**Performing**". Also added that Norming involves the development of "**unwritten rules**" / **norms** specifically (per WK5a slide).

B3 — 10 Project Management Competencies (PMBOK 2012 & Hartley 2018, WK1a)

The lecture explicitly lists them as **Project [X] Management** competencies — keep the word "Project" in mind:

#	Competency	One-line summary (per lecture)
1	Project Stakeholder Management	Identify, inform, engage and manage all project stakeholders throughout the project.
2	Project Scope Management	Ensure all work required by the project — and only this work — is included.
3	Project Time Management	Ensure the project is completed in a timely manner.
4	Project Cost Management	Ensure the project is completed within the approved budget.
5	Project Quality Management	Ensure the project satisfies the needs (standards, definitions, specifications, use) for which it was undertaken.
6	Project Human Resource Management	Make effective use of all people involved in the project.
7	Project Communications Management	Ensure all communication is complete, timely, appropriate & targeted.
8	Project Risk Management	Ensure project risks are clearly identified, comprehensively analysed and responded to proactively.
9	Project Procurement Management	Acquire all external goods & services for the project.
10	Project Integration Management	Ensure all aspects of the project are effectively coordinated and managed.

⚠ **Correction:** Original was substantively correct — added the consistent "**Project [X] Management**" naming convention used in the slide and the formal one-line descriptors as they appear in the lecture.

B4 — Important considerations for good communication; choose 2 stakeholders and discuss barriers

Considerations (per WK5b, "Planning communications management" and "Managing project communications"):

- Communication is essential to project success — 75–90% of project time is spent communicating (writing, reading, talking, listening).
- Information should be **clear, concise, accurate, timely, targeted, appropriate and complete**, and provided **in the right format, at the right time, to the right audience, with the right impact**.
- Distinguish four types of input: **Fact** (indisputable truth), **Fantasy** (someone's opinion/interpretation), **Folklore** (rumour, gossip, hearsay), **Feelings** (intuition, ego, emotion).
- The **Communication Process Model** (Belch & Belch 2004) describes communication as the passing of information between **sender** and **receiver** through encoding, message, channel, decoding — with **noise** capable of disrupting any step.

Example: Project Manager (sender) → Contractor (receiver).

- The contractor may **decode incorrectly** — different attitudes, experiences and beliefs produce different "cognitive views" of the message.
- The scope may have changed and the contractor has **not been informed**.

Example: Project Manager → Sponsor.

- The sponsor may receive **filtered or incomplete reports**, leading to misaligned expectations on cost, schedule or scope.

Common communication barriers (WK5b):

- Lack of client involvement
- Poorly informed stakeholders
- Lack of meetings, or too many meetings with no action
- Lack of reporting requirements
- Poor and incomplete documentation
- Frequent scope change
- Changing project personnel
- Lack of auditing the project to identify lessons learned

⚠ **Correction:** Original was on the right track but missing the formal **Communication Process Model** terminology (sender / encode / message / channel / decode / receiver / noise) and the **Fact / Fantasy / Folklore / Feelings** classification — both are specifically lectured in WK5b.

B5 — Why stakeholder management is important + 4 (actually 5) stakeholder strategies

Why important (WK2b):

- Stakeholders affect the project's outcome — they can support or hinder it.
- They cannot be ignored — they **will** impact ultimate success.
- They must be identified prior to commencement and throughout the project.
- Their information needs must be managed throughout the project lifecycle.
- They bring diversity of ideas, contributions, support and challenges.
- Poor stakeholder management risks projects going over time, over budget, and failing to meet targets.
- Good stakeholder management also benefits future business — recommendations and re-hire opportunities.

Five response strategies to stakeholder pressures in global projects (Aaltonen et al. 2009, WK2b):

1. **Adaptation strategy** — Obeying the demands and rules presented by stakeholders; adjusting to stakeholder pressures so the planned objectives can still be achieved.
2. **Compromising strategy** — Negotiating with stakeholders; opening dialogue, listening to their requests, and offering compensations where appropriate (**caveat: question the ethical implications**).
3. **Avoidance strategy** — Loosening attachment to stakeholders and their claims; transferring responsibility for dealing with the claims to others in the project network (**caveat: is this ethical?**).
4. **Dismissal strategy** — Ignoring stakeholder demands — i.e. not considering stakeholder requirements when implementing the project stages.
5. **Influence strategy** — Dealing proactively with the demands from stakeholders; creating and communicating value to stakeholders while building relationship with them.

⚠ **Correction:** Original called this "4 stakeholder management strategies" but the source (Aaltonen et al. 2009, lectured in WK2b) defines **5**. Also corrected the typo "*loosing attachment*" → "loosening attachment", and added that Adaptation = "obey & adjust" while Influence = "deal proactively + create value" (the original answer's wording was slightly off).

B6 — Life cycle thinking and why it is important to project management

Life cycle thinking (LCT) is a strategy that takes into account *all the impacts (economic, social and environmental — i.e. TBL) that a product or service will have throughout its life cycle*. It enables project managers to see the lasting impact of their decisions on TBL.

The "**cradle-to-grave**" life cycle (Mont et al. 2007, lectured in WK1b) covers:

- Raw materials → Resource extraction
- Manufacture → Production
- Distribution
- Usage / Consumption
- Waste disposal (and any resulting pollution)

Project management (Hartley 2018, WK1a) — "the application of knowledge, skills, tools and techniques to project activities to meet project requirements. Achieved through the application and integration of the project management processes of **initiating, planning, executing, monitoring & controlling, and closing**." (Note: PMBOK uses 5 process groups; Hartley collapses these to the 4 lifecycle stages.)

Life cycle impact assessment: *an impact is a positive or negative result of an effect of a product, process and/or activity, including all social, economic and environmental consequences and implications.*

Why it matters:

- It is consistent with TBL and forces project managers to consider the *whole* product life — not just construction.
- It enables long-term decision-making and identifies hidden environmental or social costs that traditional financial analysis misses.
- It builds positive public image, intellectual property and long-term returns.

⚠ **Correction:** Original answer was mostly correct. Added the precise stages of the "cradle-to-grave" cycle from the WK1b slide. Also clarified the PMBOK 5 process groups (initiating, planning, executing, monitoring & controlling, **closing**) — the original answer omitted "monitoring & controlling" and "initiating".

B7 — Stages of quality management (WK4b)

The lecture title is "**The road to quality**" — four stages:

1. **Quality Planning** — All mandated quality standards, operational definitions and business requirements relevant to the project are clearly identified and agreed, and ensured to be achievable and measurable throughout the project. **Quality is planned in from the start; not simply inspected in later.**
2. **Quality Assurance (QA)** — A declaration / guarantee that overall project performance (input, process & output) is evaluated on a regular basis to give all stakeholders the confidence that relevant quality standards will be satisfied. Tools include the quality management plan, regular meetings, scheduled/random quality audits, ISO standards (9000 for quality, 14000 for environmental management, 26000 for social responsibility).
3. **Quality Control (QC)** — Monitors specific task and project results to identify, measure and eliminate causes of unsatisfactory performance, while ensuring quality compliance is demonstrated and achieved. *Control implies measurement.* Tools: peer reviews, physical inspection, control charts, scatter diagrams, checklists, Pareto diagrams, statistical sampling, flowcharts, cause-and-effect (fishbone) diagrams, trend analysis.
4. **Quality (Continuous) Improvement** — A culture, commitment and ownership of *what* the project is delivering and *how well* it is being delivered. Innovation, risk-taking and efficiencies all support this. Tools: regular performance reporting, meetings/debriefs, decision gates, walkthroughs, scenario analysis, evaluation reports, suggestion boxes, user feedback.

⚠ **Correction:** Original answer was on point. Added the QA tools (ISO standards 9000/14000/26000) and QC tools (control charts, Pareto, fishbone, etc.) which are explicitly lectured. Also added the key phrase from WK4b: "**quality is planned in from the start, not randomly inspected in.**"

B8 — PERT: definition, advantages and disadvantages (WK3b)

PERT (Program Evaluation and Review Technique) is a *statistical / network-analysis* tool used to analyse and represent the tasks involved in completing a project. It illustrates: project logic and how it is tied together; relationships between tasks; flow of work; the **critical path**; potential bottlenecks; how each task is required for project completion.

The 3-point estimate (PERT, Beta distribution, PMBOK 2013):

$$Te = (To + 4Tm + Tp) / 6$$

where To = optimistic, Tm = most probable, Tp = pessimistic time.

Example from slide: To=5, Tm=10, Tp=30 → Te = 12.5 days.

Advantages (WK3b — "PERT network diagram: Advantages & disadvantages"):

- Excellent visual & interactive graphic to demonstrate the schedule
- Participative decision-making
- Joint risk identification & response strategy
- Negotiated concessions
- Improved team ownership
- Shows the **critical path**
- Eliminates idle time

Disadvantages:

- Difficult to read if the project is large
- **No timeline** (unlike Gantt)
- Difficult to monitor & report performance
- Not always easy to understand
- Limited amount of information that can be displayed

⚠ **Correction:** Original answer was correct on most items but missed that PERT also includes "**negotiated concessions**" and "**eliminates idle time**" as advantages. The 3-point estimate formula was not in the original answer — added for completeness.

B9 (first) — Balanced Scorecard (BSC) for projects (Stewart 2001, WK4b)

The BSC is a **performance measurement framework** that includes strategic performance metrics to give project managers and key stakeholders a more "balanced" view of project performance.

Four perspectives:

- **Customer perspective** — Customer satisfaction, economic value added, intended objectives.
- **Financial perspective** — Within budget; variance between original and final budget; project costs compared to industry standards.
- **Project / Internal perspective** — Team satisfaction, resource management.
- **Growth & Innovation perspective** — Best practices identified, ongoing improvement, innovative-ideas rate, TBL, life cycle.

BSC across the 4 stages (per slide):

- **Stage 1 (Concept)** — Initial measurement to establish a baseline for project planning.
- **Stage 2 (Planning)** — BSC benchmarks are included in the overall project plan.
- **Stage 3 (Execution)** — BSC measurements are implemented and initial benchmarks are used to compare and improve BSC outcomes.
- **Stage 4 (Finalisation)** — BSC measurements are reviewed and documented in the final report to support best practices and lessons learned.

⚠ **Correction:** Original answer used informal labels (e.g. "growth & innovative"). Per Stewart 2001 (WK4b slide) it is "**growth & innovation**", and the official ordering is *Customer / Financial / Internal / Growth & Innovation*. The four life-cycle applications were also paraphrased — replaced with the slide-text definitions.

B9 (second, numbered as 9 in original — actually Q10) — Contracts: legally-binding requirements and 3 types (WK6b)

A contract is a legally binding agreement between two or more parties to act, or refrain from acting, in a particular way — creating legal rights and obligations enforceable in law.

Essential elements (minimum 7) of a legally binding contract:

1. **Offer** (capable of being accepted)
2. **Acceptance** (capable of being communicated)
3. **Intention / Consent** of each party to be legally bound (freely given)
4. **Consideration** (something is given back for the promise, e.g. money)
5. **Mutuality** (neither party is disadvantaged)
6. **Capacity** (both parties are not under any duress)
7. **Legality** (it is not an illegal activity)

Plus: inclusion of both express & implied terms, conditions & warranties, and provision for breach, remedy & other non-performance redress.

Three types of contracts:

Type	Description	Customer risks	Contractor risks
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Fixed-price	Delivery of a well-defined product (goods/service) for a fixed price	At mercy of sole-source contractor; incomplete specs lead to contract changes; contractor may use cheaper materials	Requires careful cost & schedule estimation; cost growth → unprofitable project
Cost-reimbursable	Seller is paid for actual costs incurred plus a fee representing seller's profit	Final cost unknown; sole-source contractor; poor specification leads to more costs & contract changes	Fee percentage declines as costs rise; rising costs damage relationship with customer
Time & Materials (T&M)	Contains features of both fixed-fee and cost-plus contracts	No specification; open-ended; sole-source contractor	Fixed rates; no defined tasks

⚠ **Correction:** Original answer listed "**offer, acceptance, consideration, capacity, legality, mutuality, consent**" (7 items). Lecture (WK6b) confirms 7 essential elements but the precise lecture wording uses "**Intention / Consent of each party to be legally bound (freely given)**" rather than just "consent". Also added that the lecture explicitly requires *inclusion of both express & implied terms, conditions & warranties, and provision for breach & remedy*.

B10 — Budgeting techniques (WK4a)

Per WK4a, "Approaches to budgets":

- **Traditional** — Previous year's level of performance is the foundation for next year's figures.
- **Zero-based** — Ignores previous results; each activity and outlay is justified. Each activity starts with zero spending.
- **Program** — Activities are grouped together for projecting costs generated by each program or major activity.
- **Top-down** — Based on pooling the knowledge of senior managers and past results. Project costs are estimated and then passed to lower-level managers who continue the breakdown into further estimates.
- **Bottom-up** — Individual task budgets are estimated in detail by the people directly responsible for doing or managing the work. Estimates are aggregated to give the total project cost.

Two perspectives on budgets (WK4a):

- *Advantages:* improved decision-making; effective cost control; more reliable profits; assess operational financial requirements.
- *Disadvantages:* top-to-bottom support difficult to get; planning delays action; often "locked" and inflexible; restrictive of innovation & change.

⚠ **Correction:** Original answer was correct; only stylistic tightening required.

B11 — Resolving disputes between stakeholders / Five conflict-management approaches (WK5a)

Per WK5a, the **five approaches to managing conflict** (with assertion/cooperation axes):

#	Strategy	Assertion	Cooperation	Outcome	When effective
1	Avoiding	Low	Low	Lose-lose	Cannot possibly win; issue is minor; confrontation may cause more damage; time-out needed; inequitable balance of power.
2	Competing / Forcing	High	Low	Win-lose	You know you are right; stakes too high to lose; quick decisive action needed; unpopular decisions must be made; show of force required.
3	Accommodating	Low	High	Lose-win	Other's evidence is more compelling; peace/goodwill more important; want to create tactical advantage by conceding; acknowledge weakness in own position; want to avoid damaging relationship.
4	Collaborating	High	High	Win-win	Want to build alliance & relationship; need enduring commitment; want to consolidate different perspectives; need optimal outcome without sacrificing own.
5	Compromising	Mid	Mid	Partial-win/Partial-win	Outcomes only moderately important; no other option works; balance of power is even; gesture of "moving forward" required; a decision is needed.

⚠ **Correction:** Original answer used "**Forcing**" as a separate name — per the WK5a slide the official label is "**Competing / Forcing**". Original also omitted the "when effective" criteria specified in the slides. Strategy #1 (Avoiding) corresponds to lose-lose — added.

B12 — Why scope management is important + what a scope management plan should include (WK3a)

A scope management plan documents *"...how the project scope will be defined, validated and controlled"* (PMBOK 2013). It:

- Establishes the **direction and guidance parameters** on how the scope (project or product/service based) will be managed.
- Provides a **formal mechanism** to limit, assess and authorise changes on a consistent and transparent basis.

Defining the scope includes:

- What is required (**inclusions**) and what isn't required (**exclusions**)
- Establishes a **scope baseline** for comparisons and updates
- Forms the foundation of the project plan
- Investigates if **expectation meets capability** — *expectations don't always match capability*
- Identifies project deliverables, results and benefits

Why important — Controlling the scope (WK3a):

- **Scope will always change over time** ("scope creep"); other words for scope creep are innovation, continuous improvement, client change of mind, or just poor management.
- Scope control requires a **written process with formal approval**.
- Proposed changes should be assessed against **all other project variables** (time, cost, risk, contracts, quality, TBL) and approved by key stakeholders.
- Implemented changes must produce **updated project plans and related documents**.

Causes of scope creep (WK5b): poor initial requirements; unanswered questions on deliverables; lack of stakeholder involvement; evolving "exceeding expectations" mentality; discovery of new "solutions"; ineffective PM; external environmental factors.

⚠ **Correction:** Original answer was largely correct but missed the formal **scope-creep causes** taught in WK5b and the precise PMBOK definition.

B13 — Key information in the project charter (WK3a)

Example of charter format (WK3a slide):

- Project **title**
- Project **start date**
- Project **finish date**
- **Key stakeholders**
- **Business case** supporting the project (e.g. solving a problem or pursuing an opportunity)
- **Deliverables / project goals**
- **Budget information**
- **Foreseeable risks**
- **TBL and life-cycle thinking**

Tangible benefits of a good charter: nominates key stakeholders; agrees deliverables/expectations; builds commitment and conveys capability; documents agreement; identifies direction and requirements; defines baseline; authorises scheduling; facilitates management; pre-empts scope changes.

Latent dangers of a poor charter: imprecise language; inaccurate estimates; lack of detail / ignored specifications; inability to close out; misleading information; variation disputes; inconsistency with quality specs; schedule delays; unavailability of resources.

(Charter content vs. Project Scope Statement, WK3a): the Scope Statement extends the charter with project justification, detailed objectives/characteristics/requirements, milestones, assumptions, success criteria — using the **SMART framework** (Specific, Measurable, Achievable, Realistic, Time-framed).

⚠ **Correction:** Original answer's "Problem / Goal / Objectives / Success criteria / Risks" was a high-level summary; the lecture template explicitly lists the 9 fields above. Added the **SMART framework** for objective setting which the original answer omitted.

B14 — Key characteristics of a highly performing project team (WK5a)

From WK5a "Developing the project team" and "What effective team members need":

- **Clear, communicated, recognised and accepted long-term goals and objectives** — every member understands where the team is headed.
- **Healthy team culture** with mutual appreciation of members' individual technical and broad skills.
- **Defined, communicated and accepted roles** — everyone knows what they are supposed to do and how it supports the team.
- **Tolerance for calculated risk and unqualified opportunities for success.**
- **Explicit, discussed and endorsed procedures.**
- **Open, honest, continuous communication and constructive feedback.**
- **Supported leadership** — trust and support each other and the leader.
- **Ability to identify key issues, solve problems and implement solutions.**
- **Self-direction** — able to work without ongoing supervision.

Indicators of team cohesion (WK5a):

- Positive: cooperative climate, trust & open support, clear expectations, exchange of ideas, directions followed, disputes resolved, open communication, ongoing constructive feedback.
- Negative: delays with decisions, withholding information, competing agendas, directions challenged, under-performance, suspicion/animosity/distrust, limited feedback, subgroups forming.

⚠ **Correction:** Original list was reasonable but informal. Replaced with the explicit lecture-slide wording from WK5a. Added the positive/negative cohesion indicators as supporting examples.

B15 — Information in the project close-out report (WK7)

The close-out (finalisation) report — and the activities surrounding it (WK7, "Closing down the project"):

Getting client acceptance (100%) — detailed completion checklist; satisfaction surveys; defects warranty; written (formal) acceptance procedure; specification conformance inspection & testing; reviews & evaluations;

walkthroughs; final payment.

Debrief the stakeholders — was the project delivered on time/budget/scope/specifications? Were all variations and scope changes addressed competently? Were TBL and life cycle included? Were communication channels effective? Was reporting documentation accurate? Was the project managed professionally?

Project performance review — final "kick-out" meeting; concise project history; explanation of variances; documented successes & failures; evaluation of lessons; recommendations for future projects; identify processes requiring amendment; assess administrative support; review project performance; evaluate suitability of tools/processes; critically assess organisational structure; review accuracy of estimates; evaluate team performance.

Learning the lessons — missed deadlines, inaccurate estimates, schedule revisions, inspection/testing failures, poor contractor performance, lack of team cohesion, cost over-runs, unauthorised scope changes, poor quality (incl. TBL) requiring rework, unresolved conflict.

KRAC framework for process improvement:

- **K — Keep** (if it works, keep it)
- **R — Remove** (if it doesn't work, remove it)
- **A — Add** (was something missing)
- **C — Change** (what can change)

Archiving the project file — single source location; all supporting information; close-out checklist; required approvals & sign-off; secure the complete project file.

⚠ **Correction:** Original answer was very brief. Expanded with the formal close-out activities lectured in WK7, including the **KRAC framework**, which the original omitted.

B16 — Procurement management planning and its benefits (WK6b)

Definition. Procurement management = *"the processes necessary to purchase or acquire products, services or results needed from outside the project team."* A **value-adding function** that focuses on the efficient attainment of goods/services/results delivering specific, measured benefits, while supporting the changing scope/goals/objectives of the project. **TBL and life-cycle thinking are at the core of procurement.**

Procurement drivers (WK6b):

- Supply the project with goods & services as specified and required
- Improved relationships with key suppliers
- Balance output with both value & quality
- Increase accountability in the supply chain
- Encourage greater access to innovative technology, premium materials, workplace competencies and expertise
- TBL and life-cycle thinking as the project core

The "8 Rs" of procurement (WK6b):

1. Right materials (consider disposal issues)
2. Right price
3. Right quantity
4. Right quality
5. Right source
6. Right reasons
7. Right time
8. Right return

Scope of work (SOW) = what is required (the *what*).

Specification = the technical detail of how it must behave / function (the *how*).

(WK6b)

⚠ **Correction:** Original answer was correct in spirit but missed the **8 Rs of procurement**, the explicit SOW-vs-Specification distinction, and that **TBL/life-cycle thinking is at the core of procurement** (WK6b emphasises this throughout). Added.

B17 — Team conflict can have positive and negative outcomes (WK5a)

Per WK5a, "Balancing conflict":

Positive outcomes:

- Exploration of new ideas
- Consideration of other people's perspectives
- Adjustments / modifications made
- Clarification of different positions / interests
- Postponed decisions
- Time to reconsider, clarify and communicate a proposal

Negative outcomes:

- The breakdown in communication
- Increased hostility
- The cessation of work on the project
- Legal action taken for contract breaches
- Project personnel being replaced

⚠ **Correction:** Original answer was correct. Minor wording adjustments to match the WK5a slide text exactly (e.g. "**postponed decisions**" and "**clarification of different positions/interests**" are explicit slide items).

B18 — 5 ways to terminate a project

The original answer cited (from Meredith & Mantel, standard PM textbook):

- **Extinction** — formal agreement to terminate (whether successful or not).
- **Murder** — a sudden, "silent" / unilateral decision to kill the project.
- **Inclusion** — a successful project is institutionalised inside the organisation as a permanent function.
- **Integration** — elements of a successful project are absorbed / redistributed to other parts of the organisation.
- **Starvation** — the project is gradually starved through budget cuts, resource reallocation, or changing priorities.

⚠ **Correction:** This taxonomy is **not directly covered in the 20 lecture slide decks** indexed (Week 1–12, excluding Week 8). It comes from the broader PM literature (Meredith & Mantel, *Project Management: A Managerial Approach*). The original answer is **technically correct** but be aware that the lecturer's focus is on the **Finalisation / Closing Down** activities in WK7 (client acceptance, debrief, performance review, KRAC, archiving). If exam answer requires explicit slide content, refer to WK7 instead. The names above are retained because the question was phrased that way in the original docx — but flagged as "**outside lecture coverage**".

Section C — Ethics

C1 — Utilitarianism, deontology and virtue-based ethics (WK9b)

The lecture identifies these as the three **main ethics theories**:

1. Utilitarian / Consequences-based theory.

An action is judged ethical or unethical based on its consequences / outcome.

- Founders: Jeremy Bentham (1748–1832) and John Stuart Mill (19th c.).
- Utilitarian thinking: "*the end justifies the means.*"
- Forward-looking; an action is right if it produces the **greatest good for the greatest number** / maximises overall utility — the most common form of **consequentialism**.
- Basic principles: (a) maximise benefits, (b) minimise costs.
- Dominates Western business thinking — e.g. cost-benefit analysis.
- **Issues:** minority rights at risk; who decides what "good" is; what counts as a consequence; how far into the future should consequences be calculated.
- *Example from lecture:* a pharmaceutical company releases a drug with minor side effects; the drug benefits a large population more than the small number troubled by side effects.
- *Ford Pinto case* (1970) — Ford did a cost/benefit analysis and chose not to fix a fuel-tank design flaw because the cost (137M) *exceeded the calculated benefit* (49.5 M). A classic utilitarian-thinking failure.

2. Deontological / Duty-based theory.

Actions are judged ethical standing alone and without regard to consequences.

- From Greek *deon* = duty.
- Most famous deontologist: **Immanuel Kant (1724–1804)**.
- Backward-looking; do the right thing simply because it is the right thing to do — **means rather than end**.
- Kant proposed three formulations of the **categorical imperative**:
 1. *Act as if, through your actions, you were making universal law for everyone to follow* — e.g. don't bribe contractors, don't mislead customers.
 2. *Always treat any human being (self included) as an end in themselves, never merely as a means to an end* — e.g. respect employee/customer safety and dignity.
 3. *Act as if you were a member of a community of fellow moral legislators who are ends in themselves* — e.g. do not force your will upon another.
- **Typical duties**: fidelity, gratitude, justice, beneficence, self-improvement, non-maleficence.
- *Example*: a manager turning down a promotion that requires forgery because it conflicts with his principles.

3. Virtue-based theory.

The focus is on the individual, not on the action.

- Ongoing quest for goodness and virtue — individuals should seek to live a virtuous life.
- A "good person" does the right thing.
- Roots in Aristotle's eudaimonia (flourishing): virtue lies in the **mean** between excess and deficiency; highest form of mortal activity is contemplation; perfection through moderation of reason and desire.

⚠ **Correction**: Original answer's coverage was correct in essence but used "*the man turning down the promotion which required him to commit forgery*" without citing Kant's three formulations. Also missed the **Ford Pinto case** as the lecture's flagship example of utilitarian failure. Added.

C2 — 5 (actually 8) Key principles in the ethics code for engineers (WK10b)

The lecture (WK10b, "Roles of codes of ethics for engineers" — Schinzinger et al. 2010) lists 8 roles of codes of ethics for engineers:

1. **Serving and protecting the public** — Engineers have advanced expertise the public lacks. They stand in a **fiduciary relationship** with the public; trust and trustworthiness are essential.
2. **Guidance** — Codes provide helpful guidance by articulating the main obligations of engineers.
3. **Inspiration** — Codes express the profession's collective commitment to ethics, providing positive stimulus for ethical conduct.
4. **Shared standards** — Diversity of moral viewpoints among engineers makes it essential that the profession establishes explicit (minimum but high) standards. The public is assured of a standard of excellence; professionals get a fair playing field for clients.
5. **Support for responsible professionals** — A publicly proclaimed code allows an engineer under pressure to act unethically to say: "*I am bound by the code of ethics of my profession, which states that...*"
6. **Education and mutual understanding** — Codes encourage shared understanding among professionals, the public and government bodies about engineers' moral responsibilities.

7. **Deterrence and discipline** — Codes serve as the formal basis for investigating unethical conduct. Professional societies do suspend or expel members whose conduct is proven unethical.
8. **Contributing to the profession's image** — Codes present a positive image to the public of an ethically committed profession. Reputation is essential in sustaining public trust.

Engineers Australia — the 4 key principles (also mentioned at end of original docx, and consistent with WK10b on Code of Ethics):

- Demonstrate **integrity**
- Practice **competently**
- Exercise **leadership**
- Promote **sustainability**

And the *moral commitments* per WK10b: **honesty and integrity, impartiality, fairness and equity, plus dedication to the protection of public health, safety and welfare.**

⚠ **Correction:** Original question asked for **5 principles** but the answer listed **8** (Schinzinger et al. 2010, WK10b) — so the question itself is mis-numbered. The correct lecture content has **8 roles** for the code of ethics (lectured in WK10b). The original answer's last block ("5 key principles in the ethics for engineers — honesty, impartiality, fairness, equity, dedicated to protection of public health, safety and welfare") is the **Engineers Australia code's underlying moral commitments**, *not* the same as Schinzinger's 8 roles. **Both are valid but they answer different questions.** Document this distinction.

C3 — Benefits of managing a project ethically (WK11b, WK12a)

1. **Cultivate strong teamwork and productivity** — Ongoing attention and dialogue regarding values builds openness and integrity. Strong alignment between team values and organisational values produces stronger motivation and performance. Promotes a positive work environment and respects diversity.
2. **Support the team's growth and meaning** — Team members feel confident they can admit and deal with whatever issues come their way; they are supported when raising ethical concerns.
3. **Helps manage values associated with management systems** — e.g. Total Quality Management depends on operating values like trust, performance and honest feedback.
4. **Promote a strong public image** — An organisation that gives regular attention to ethics earns a strong positive public image and reputation.
5. **Long-term sustainability** — Ethical management aligns naturally with TBL (financial, social, environmental responsibility) and life-cycle thinking. Reputational capital, public trust, and reduced risk of legal/regulatory issues all translate to long-term commercial value.

⚠ **Correction:** Original list was correct; added the explicit **TBL / sustainability link** (WK11b–WK12a emphasises that ethics + TBL is a complementary pairing — sustainability is one of the key themes the lecturer summarises in WK12a "In conclusion").

C4 — 5 key stakeholder groups and their ethical values

- 1. Client (the initiator)** — the person/entity wanting the project to exist; ensures the project is aligned with the strategic needs and goals of the organisation; will not want the project to be immoral or have a negative outcome.
- 2. Sponsor (funding party)** — has **ultimate accountability and responsibility** for the project; holds significant influence; obtains funding; monitors costs/benefits/risks; ultimately has the final word on whether a project is to be deemed ethical and carried out.
- 3. Project Steering Group** — the senior management team responsible for adoption, deployment and uniform application of the project management framework; **ensures everyone, including minorities, has a voice**; balances competing priorities; approves commencement and sign-off at each phase.
- 4. Project Manager** — appointed to manage the project; **leads by example**; ensures the team performs in an ethical manner; communicates shared ethical values; rewards ethical behaviour and disciplines unethical behaviour; **protects whistle-blowers**.
- 5. Project Team Members** — subject-matter experts/personnel performing all required work; **complete their tasks ethically to the required standards**; communicate emergent issues and risks honestly; participate in performance/progress meetings.

⚠ **Correction:** Original answer correctly identified the 5 groups (per WK2b's stakeholder hierarchy). Added the sponsor's role of having "**ultimate accountability**" and the PM's role of **leading by example and protecting whistle-blowers** (both per WK12a "Attributes of an ethical Manager/Project Manager/Leader").

C5 — 7 Stages of the Ethical Cycle (Van de Poel & Royakkers 2007, WK10b)

- 1. Case** — the cycle starts with a particular case of moral complication.
- 2. Moral problem statement** — *precise / specific formulation*. Clearly state what the problem is, for whom it is a problem, and the moral nature of the problem.
- 3. Problem analysis** — *clear identification of relevant elements*:
(a) stakeholders; (b) moral values involved; (c) relevant facts.
- 4. Options for action** — *all possible options should be identified* — **thinking creatively is essential**.
- 5. Ethical judgement** — *thorough application of moral concepts, ethical theories, models and frameworks* (utilitarian, deontological, virtue, code of ethics, etc.).
- 6. Reflection** — *reflect on the outcome of the previous stage* — does the chosen action stand up to scrutiny?
- 7. Morally accepted action(s)** — actions decided upon are presented and implemented; chosen to solve the case / ethical dilemma.

⚠ **Correction:** Original answer was substantively correct. Added the lecture-specific qualifiers — "(a) stakeholders; (b) moral values; (c) relevant facts" under step 3, and the "**thinking creatively is essential**" note in step 4 — both lifted from the WK10b slide.

C6 — Characteristics of an ethical manager (WK12a)

Ethical managers / project managers / leaders provide a good role model by **being ethical and honest at all times**, obeying both wide loyalty and integrity principles:

- **Telling the truth**
- **Admitting failure**
- **Communicating shared ethical values to employees**
- **Rewarding employees who behave ethically and punishing those who do not**
- **Protecting employees (whistle-blowers)** who bring to light unethical behaviours or raise ethical issues
- **Lead by example**

Additionally (WK12a "Attributes of an ethical Manager continues..."):

- **Establish codes of ethics and decision rules**
- **Hire individuals with high ethical standards**
- **Have all levels of management continually reaffirm the importance of the code of ethics** and the organisation's commitment to the code
- **Publicly reprimand and consistently discipline those who break the code**
- **Provide ethics training**
- **Conduct independent social audits**
- **Provide support for individuals facing ethical dilemmas**

Transformational vs Transactional leadership (WK12a):

- *Transactional leaders* — Use social exchanges; link performance to rewards; ensure efficiency.
- *Transformational leaders* — Stimulate and inspire followers to transcend self-interest for the good of the organisation; create vision; communicate vision; **model the vision** ("walk the talk"); build commitment.
- Research shows transformational leaders implement a more consistent ethical approach than transactional leaders.

⚠ **Correction:** Original list was correct. Added: (i) **publicly reprimand & consistently discipline** code-breakers (a specific WK12a slide point), (ii) **conduct independent social audits**, (iii) the contrast between **transactional vs transformational leadership** which the lecturer presents as part of ethical leadership.

C7 — Role of ethical engineers to the public (WK10b)

Engineering has a **direct impact on the quality of life for all people**. Engineers must:

- Maintain **honesty and integrity, impartiality, fairness and equity**, and be dedicated to the **protection of public health, safety and welfare**.
- Perform under a standard of **professional behaviour** and adhere to the highest principles of ethical conduct.
- Be guided by frameworks for ethics (codes of ethics).

The role of an engineering code of ethics (Schinzinger et al. 2010):

- (a) protect public health, safety, welfare;
- (b) share their understanding of moral responsibilities amongst engineers;
- (c) maintain integrity and trust throughout careers.

Engineers Australia Code of Ethics — the 4 key principles:

- Demonstrate **integrity**
- Practice **competently**
- Exercise **leadership**
- Promote **sustainability**

⚠ **Correction:** Original answer was correct. Slight rewording to align with the WK10b slide text.

C8 — Tests a project manager can perform to determine if a task is ethical (WK10a)

Per WK10a slide "How (2)?", a project manager can apply the following **nine tests**:

#	Test	Question
a	Legal	Does the action violate the law?
b	Duties	Does it contradict widely accepted moral obligations (fidelity, gratitude, justice, non-maleficence, beneficence)?
c	Consequences	Will it result in major damages to people or organisations?
d	Utilitarian	Is there a satisfactory alternative action that produces equal or greater benefits to the parties affected?
e	Special obligations	Does it violate any other special duties (e.g. protect consumers and provide safe products)?
f	Motives	Is the intent of the contemplated action harmful?
g	Rights	Does it infringe upon the rights of stakeholders in any way?
h	Justice	Does it leave another person less well off?
i	Sustainability	Does it impact negatively on future generations?

⚠ **Correction:** Original answer was correct. Re-formatted into a table and added the lecture's parenthetical examples (e.g. *fidelity, gratitude, justice, non-maleficence, beneficence* for the Duties test) for clarity.

Additional Lecture Content Worth Knowing (not in original docx)

The following major topics appear in lectures but are absent from the original 5505.docx . Including them here for exam coverage.

Project selection — non-numeric models (WK2a)

- **Sacred cow** — protected by senior executive.
- **Operating necessity** — to maintain operational functionality (e.g. during a crisis).
- **Competitive necessity** — to maintain competitive advantage / match competitors.
- **Product extension** — extending existing product lines.
- **Comparative benefit** — for companies seeking multiple projects with differing benefits.

Risk management (WK6a)

Risk = the possibility of loss or injury; a situation with measured impact (positive or negative) on an outcome.

Process (iterative): Identify → Assess → Analyse → Manage → Monitor & Evaluate → Review.

5×5 priority grid: Probability × Impact = priority score (1–6 = existing controls; 8–12 = project manager intervention; 15–25 = senior management intervention).

Possible risk responses: Reject, Accept, Mitigate, Share, Transfer, Enhance (for opportunities), Exploit (for opportunities).

Tools: SWOT, PESTELG, risk registers, fishbone diagrams, lessons-learned reports, simulations, decision trees, sensitivity analysis, contingency planning, financial modelling.

Cross-cultural business ethics (WK11b)

6 universal moral values for corporate codes of ethics (Schwartz 2005):

1. Trustworthiness (integrity, honesty, loyalty, transparency)
2. Respect (for human rights)
3. Responsibility (accountability, self-restraint)
4. Fairness (equity, impartiality)
5. Caring (no harm)
6. Citizenship (TBL, life-cycle thinking, obeying the law)

UN Global Compact — 10 Principles in human rights, labour, environment, anti-corruption.

4 options for a multinational facing home-host norm conflict (Beauchamp et al. 2008):

1. Follow norms of home country (patriotic).
2. Follow norms of host country ("when in Rome...").
3. Follow whichever norm is most profitable.
4. Follow whichever norm is morally best.

Moral development — Stages hierarchy model (Linstead et al. 2004 / Kohlberg, WK11a)

- **Pre-conventional morality** (Stages 1–2) — self-interest, "what's in it for me?"
- **Conventional morality** (Stages 3–4) — pleasing close associates (3); professional integrity and lawful pursuit (4).
- **Post-conventional morality** (Stages 5–7) — concern for social justice and collective well-being (TBL approach); less than 1/5 of adults reach Stage 5, and Stages 6–7 are rare.

Line-drawing approach to ethical dilemmas (Fledderman 2011, WK11a)

- Place a **positive paradigm (PP)** at one end (unambiguously morally acceptable) and a **negative paradigm (NP)** at the other end (unambiguously unacceptable).
- Place the problem **P** under consideration between them, along with hypothetical examples.
- The closer **P** is to the positive paradigm, the more morally acceptable the action.
- Caveat: subjective, legal, political and community-perception aspects must also be considered.

Negotiation — Getting to Yes (Fisher & Ury 1991, WK6b)

- Separate the people from the problem.
- Focus on interests, not positions.
- Invent options for mutual gain.
- Insist on using objective criteria.
- Develop your **BATNA** (Best Alternative to a Negotiated Agreement) and be patient.

Setting objectives — SMART (WK3a)

Specific, Measurable, Achievable, Realistic, Time-framed.

Cradle-to-Grave life cycle (WK1b, Mont et al. 2007)

Raw materials → Resource extraction → Manufacture/Production → Distribution → Usage/Consumption → Waste disposal.

Sources used for the corrections

Week	File	Key topics referenced
1a	Lecture WK1a 24-2-26.pdf	Definitions of project & project management; 10 PMBOK competencies; sustainability & TBL
1b	Lecture WK1b 26-2-26.pdf	4 project life cycle stages; cradle-to-grave; benefits of LCT
2a	WK2a 03-3-2026.pdf	Vision & strategy; project selection (numeric & non-numeric); payback, ROI, NPV
2b	WK2b 05-3-2026.pdf	Stakeholder management; 5 response strategies (Aaltonen 2009); RACI/PARIS

3a	Lecture WK3a 10-3-26.pdf	Scope management; project charter; SMART
3b	Lecture WK3b 12-3-26.pdf	WBS; PERT; 3-point estimate; Gantt; critical path
4a	WK4a 17-3-26.pdf	Budgeting approaches (traditional / zero-based / program / top-down / bottom-up); EVM
4b	WK4b 19-3-26.pdf	Quality (planning/assurance/control/improvement); BSC; sustainability indicators
5a	WK5a 24-3-26.pdf	Team development (Forming/Storming/Norming/Performing/Adjourning); conflict modes; MBTI
5b	WK5b 26-3-26.pdf	Communications management; Comm Process Model; barriers; scope creep
6a	WK6a 31-3-26.pdf	Risk management; SWOT; PESTELG; 5×5 grid; risk responses
6b	WK6b 2-4-26.pdf	Procurement; 8 Rs; contract elements; 3 contract types; negotiation
7	WK7 14-4-26.pdf	Integration management; closing down; KRAC; archiving
9a	WK9a 28-4-26.pdf	Intro to ethics
9b	WK9b 30-04-26.pdf	Utilitarian / deontological / virtue theories; Kant; Ford Pinto
10a	WK10a 5-05-26.pdf	Virtue ethics; 9 tests for ethical task
10b	WK10b 7-05-26.pdf	Ethical cycle (7 steps); code of ethics; 8 roles of codes
11a	WK11a 12-5-25.pdf	Line drawing; moral development stages
11b	WK11b 14-5-26.pdf	Cross-cultural business ethics; Schwartz's 6 values; UN Global Compact
12a	WK12a 19-5-26.pdf	Ethical management & leadership; transactional vs transformational